

RKHS Theory and the Support Vector Machine

A Mathematical Treatment of Reproducing Kernel Hilbert Spaces and Kernel SVMs

Tushar Moitra

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1 Reproducing Kernel Hilbert Spaces

1.1 Core Definitions

Let $\mathcal{X}$ be a non-empty set (our input domain, typically $\mathbb{R}^d$).

Definition 1.1 (Positive Definite Kernel). A symmetric function $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is *positive definite* (p.d.) if for all $n \in \mathbb{N}$, all $\{x_i\}_{i=1}^n \subset \mathcal{X}$, and all $\mathbf{c} \in \mathbb{R}^n$,

$$\sum_{i=1}^n \sum_{j=1}^n c_i c_j K(x_i, x_j) \geq 0.$$

K is *strictly* p.d. if equality implies $\mathbf{c} = \mathbf{0}$.

Definition 1.2 (RKHS). A Hilbert space $(\mathcal{H}, \langle \cdot, \cdot \rangle_{\mathcal{H}})$ of functions $f : \mathcal{X} \rightarrow \mathbb{R}$ is a *reproducing kernel Hilbert space* if there exists $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ such that

- (i) $K(\cdot, x) \in \mathcal{H}$ for all $x \in \mathcal{X}$;
- (ii) (*Reproducing property*) $f(x) = \langle f, K(\cdot, x) \rangle_{\mathcal{H}}$ for all $f \in \mathcal{H}$, $x \in \mathcal{X}$.

K is called the *reproducing kernel* of $\mathcal{H}$.

The reproducing property immediately gives

$$K(x, y) = \langle K(\cdot, x), K(\cdot, y) \rangle_{\mathcal{H}},$$

which shows K is symmetric and positive definite.

1.2 The Moore–Aronszajn Theorem

Theorem 1.3 (Moore–Aronszajn, 1950). *Every positive definite kernel K on $\mathcal{X}$ defines a unique RKHS $\mathcal{H}_K$ whose reproducing kernel is K .*

Sketch. Form the pre-Hilbert space

$$\mathcal{H}_0 = \text{span}\{K(\cdot, x) : x \in \mathcal{X}\}$$

with inner product

$$\left\langle \sum_i \alpha_i K(\cdot, x_i), \sum_j \beta_j K(\cdot, y_j) \right\rangle := \sum_{i,j} \alpha_i \beta_j K(x_i, y_j).$$

Positive definiteness ensures this is well-defined and non-negative. Taking the completion gives $\mathcal{H}_K$.

2 From Primal SVM to the RKHS Dual

2.1 Primal Formulation

Given training data $\{(x_i, y_i)\}_{i=1}^n$ with $y_i \in \{-1, +1\}$ and $x_i \in \mathbb{R}^d$, the soft-margin SVM primal is

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \tag{1}$$

subject to

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

The geometric margin is

$$\frac{2}{\|w\|}.$$

Maximizing margin is therefore equivalent to minimizing $\|w\|^2$.

2.2 Lagrangian and KKT Conditions

Introduce multipliers $\alpha_i \geq 0$ and $\mu_i \geq 0$.

The Lagrangian becomes

$$L_P = \frac{1}{2} \|w\|^2 + C \sum_i \xi_i - \sum_i \alpha_i [y_i(w^T x_i + b) - 1 + \xi_i] - \sum_i \mu_i \xi_i.$$

Stationarity conditions give

$$w = \sum_i \alpha_i y_i x_i, \quad (2)$$

$$\sum_i \alpha_i y_i = 0, \quad (3)$$

$$\alpha_i + \mu_i = C. \quad (4)$$

Hence

$$0 \leq \alpha_i \leq C.$$

Remark 2.1. The optimal weight vector lies in the span of the training data:

$$w \in \text{span}\{x_1, \dots, x_n\}.$$

This is the finite-dimensional precursor to the Representer Theorem.

2.3 Dual Formulation

Substituting the KKT conditions gives the Wolfe dual:

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j (x_i^T x_j) \quad (5)$$

subject to

$$0 \leq \alpha_i \leq C, \quad \sum_i \alpha_i y_i = 0.$$

Observe carefully:

The dual depends only on inner products.

This is the structural reason kernel methods work.

3 The Kernel Trick and RKHS Geometry

3.1 Feature Space Geometry

Let

$$\Phi : X \rightarrow \mathcal{H}_K$$

be the feature map induced by a kernel K .

Then

$$K(x, y) = \langle \Phi(x), \Phi(y) \rangle_{\mathcal{H}}.$$

The SVM classifier becomes

$$f(x) = \langle w, \Phi(x) \rangle_{\mathcal{H}} + b.$$

The primal optimization problem is now posed in the RKHS $\mathcal{H}_K$:

$$\min_{w \in \mathcal{H}_K, b, \xi} \frac{1}{2} \|w\|_{\mathcal{H}}^2 + C \sum_i \xi_i.$$

The derivation proceeds identically.

The optimal solution satisfies

$$w = \sum_i \alpha_i y_i \Phi(x_i).$$

Substituting into the classifier gives

$$f(x) = \sum_i \alpha_i y_i \langle \Phi(x_i), \Phi(x) \rangle_{\mathcal{H}} + b.$$

Using the reproducing property:

$$f(x) = \sum_i \alpha_i y_i K(x_i, x) + b$$

This is the kernel SVM.

3.2 Representer Theorem

Theorem 3.1 (Representer Theorem). *Every minimizer of a regularized empirical risk problem in an RKHS admits a representation*

$$f^*(\cdot) = \sum_{i=1}^n \alpha_i K(\cdot, x_i).$$

Proof sketch.

Decompose

$$f = f_{\parallel} + f_{\perp}$$

where

$$f_{\parallel} \in \text{span}\{K(\cdot, x_i)\}$$

and

$$f_{\perp} \perp K(\cdot, x_i).$$

By the reproducing property,

$$f(x_i) = \langle f, K(\cdot, x_i) \rangle_{\mathcal{H}} = \langle f_{\parallel}, K(\cdot, x_i) \rangle_{\mathcal{H}}.$$

Hence the orthogonal component does not affect training loss.

But

$$\|f\|^2 = \|f_{\parallel}\|^2 + \|f_{\perp}\|^2 \geq \|f_{\parallel}\|^2.$$

Therefore any optimizer must satisfy

$$f_{\perp} = 0.$$

Hence the solution lies entirely in the span of kernel sections.

3.3 Geometric Interpretation

The kernel defines:

- the feature geometry,
- the notion of similarity,
- the admissible function class,
- the topology of the RKHS.

The SVM then chooses the minimum-norm separator in that geometry.

Thus:

$\text{Kernel} \Rightarrow \text{Geometry} \Rightarrow \text{Optimization}$

4 Conclusion

RKHS theory provides a bridge between:

- functional analysis,
- geometry,
- optimization,
- and machine learning.

The reproducing property transforms pointwise evaluation into Hilbert geometry:

$$f(x) = \langle f, K(\cdot, x) \rangle_{\mathcal{H}}.$$

Support Vector Machines exploit this structure by solving a linear maximum-margin problem inside a potentially infinite-dimensional RKHS while only computing kernel evaluations.

The success of kernel methods therefore comes from the fact that nonlinear problems become linear geometric problems in carefully designed Hilbert spaces.